Math Spirit: Theoretical Foundations

Algorithmic Enhancements in Multi-Agent Particle Environments

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1 Introduction & Motivation

In the development of our MPE project, we encountered a critical challenge: **Sparse Rewards**. While it is tempting to manually design rewards to guide agents, this often leads to unintended "Reward Hacking" behaviors.

The core motivation for our current approach stems from a fundamental question raised during technical review:

> *"How to provide guidance to the Agent without distorting the original objective? Specifically, how to prevent agents from oscillating near a target just to exploit a distance-based reward?"*

This led us to re-examine the principles of **Policy Invariance**.

2 PBRS: Theoretical Background

To rigorously address reward hacking, we implement **Potential-Based Reward Shaping (PBRS)**, a concept formalized by **Andrew Ng, Daishi Harada, and Stuart Russell (1999)** in their seminal paper *"Policy Invariance under Reward Shaping"*.

2.1 The Ng et al. Necessary and Sufficient Condition

Ng et al. provided a mathematical guarantee for maintaining the optimal policy π^*. They proved that a shaping function $F(s, a, s')$ preserves the optimality of a policy if and only if it takes the following form:

Fundamental Theorem of PBRS

$$F(s, a, s') = \gamma\Phi(s') - \Phi(s)$$

where $\Phi : S \to \mathbb{R}$ is a real-valued potential function of states, and γ is the MDP's discount factor.

2.2 Mathematical Proof of Policy Invariance

By substituting $F(s, a, s')$ into the Bellman optimality equation, the relationship between the original Q-value (Q) and the shaped Q-value (Q') can be derived:

$$Q'(s, a) = \mathbb{E}[r + \gamma\Phi(s') - \Phi(s) + \gamma \max_{a'} Q'(s', a')] \tag{1}$$

$$Q'(s, a) + \Phi(s) = \mathbb{E}[r + \gamma(Q'(s', a') + \Phi(s'))] \tag{2}$$

Let $\hat{Q}(s, a) = Q'(s, a) + \Phi(s)$. The equation recovers the original Bellman form. Since $\Phi(s)$ is independent of the action a, it follows that:

$$\arg\max_a Q'(s, a) = \arg\max_a \hat{Q}(s, a) = \arg\max_a Q(s, a)$$

Thus, the optimal policy $\boldsymbol{\pi^*}$ remains unchanged.

3 Ongoing Enhancements

Beyond PBRS, the *Math Spirit* repository incorporates:

- **Control Variable Methods:** Reducing gradient variance in Multi-Agent training.
- **Automatic Control Principles:** Leveraging stability analysis to constrain agent behaviors.